

# Exploratory data analysis of Global Video Game Sales



(Kaggle [dataset](#))

Analysis & visualizations made with Python, by Elodie Barthout

Last update: August 27th, 2026

# Content

- Dataset description
- Data cleaning process
- General analysis
- EDA
  - Focus on global sales
  - Focus on North America sales
  - Focus on Europe sales
  - Focus on Japan sales
  - Focus on Other Regions sales
  - Differences between regions
  - Focus on the global best-selling genres
- Conclusion

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# Dataset description

This dataset was sourced from [vgchartz.com](https://vgchartz.com) & offers insights into the **video game industry** by giving information on:

- Sales (globally as well as for specific regions)
- Year of release
- Platform of release
- Genre
- Publisher

The aim of this exploratory data analysis is to understand **consumers' preferences** in terms of video games, depending on the regions: which genre is the most appreciated, which video game, which platform was it mostly sold on ect.

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# Data cleaning process

## Before cleaning:

- 16 598 entries
- 11 493 unique games
- 12 genres
- 31 platforms
- 1980-2020 year distribution

## After cleaning:

- 12 808 entries
- 8736 unique games
- 12 genres
- 29 platforms
- 1980-2016 year distribution

## Cleaning process:

- Fixing inconsistencies
- Recovering missing values when possible
- Removing
  - Duplicate records
  - Global sales < 0.05
  - Missing values

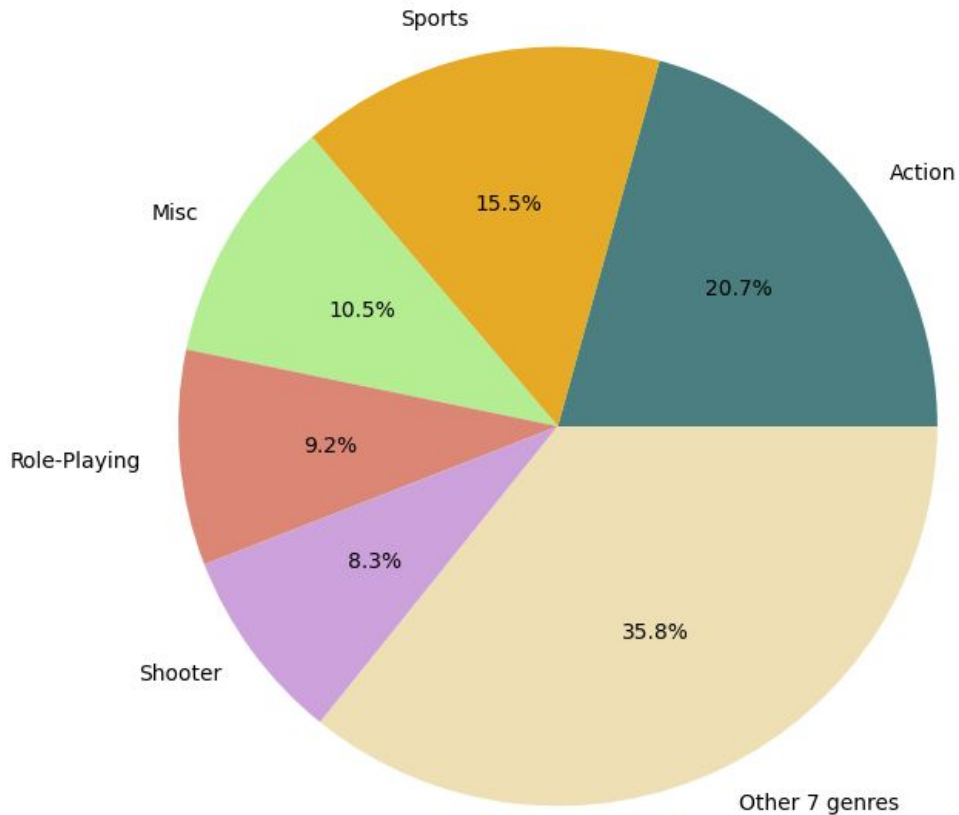
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# Genre distribution

From 1980 to 2016, most of the video games were of either action, sports, misc, role-playing or shooter genre.

The least represented genres being puzzle & strategy.



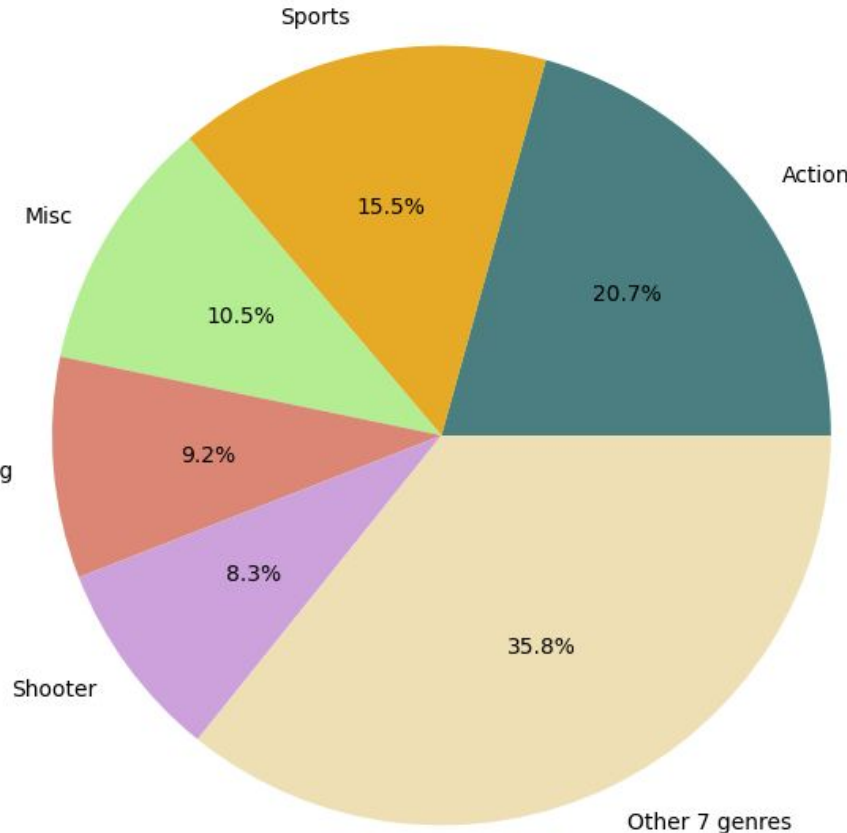


# First video games

Ice Hockey released  
in 1980 by Activision  
on the Atari 2600

4 games released on the Atari  
2600 by Activision & Atari in  
1980: Kaboom!, Defender,  
Bridge, Checkers

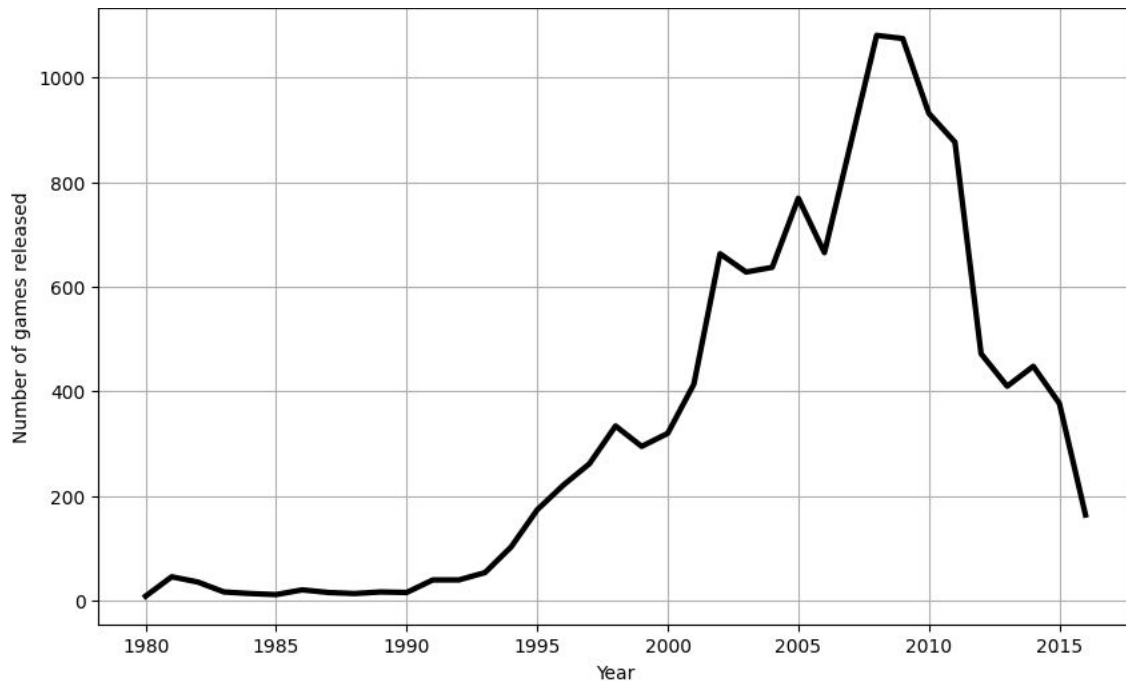
Freeway released in  
1980 by Activision on  
the Atari 2600



Dragon Warrior released  
in 1986 by Capcom on  
the NES

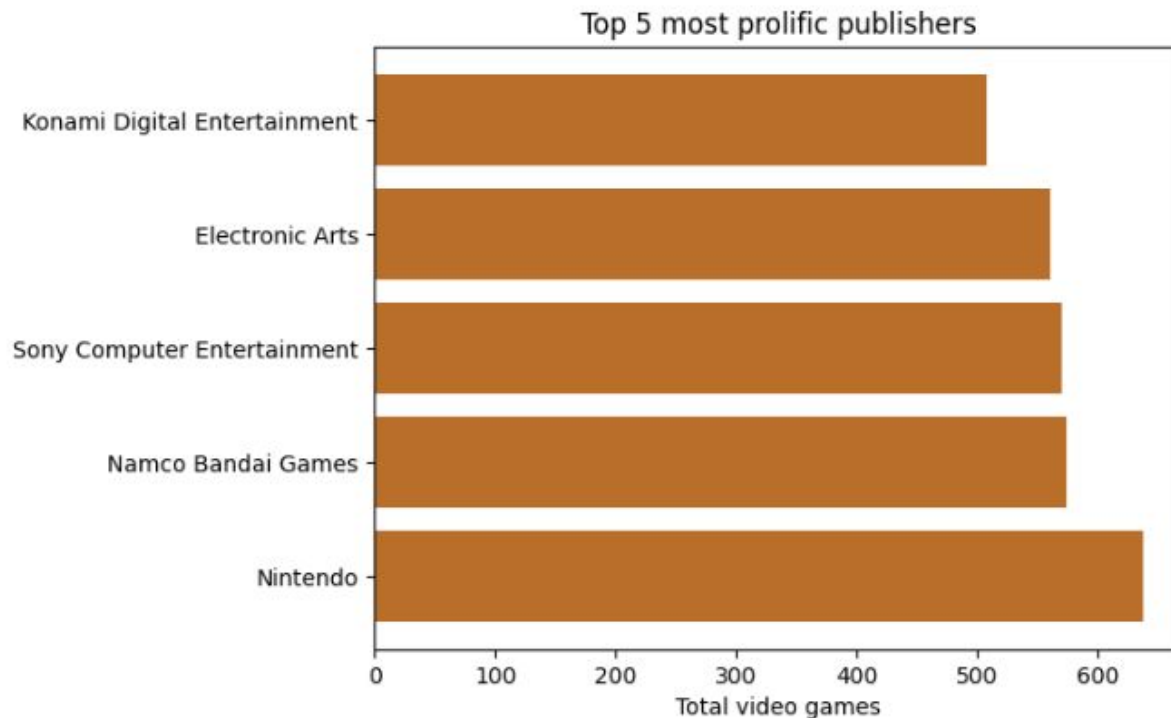
2 games released on the Atari  
2600 by Atari in 1980: Asteroids  
& Missile Command

# Number of releases over time



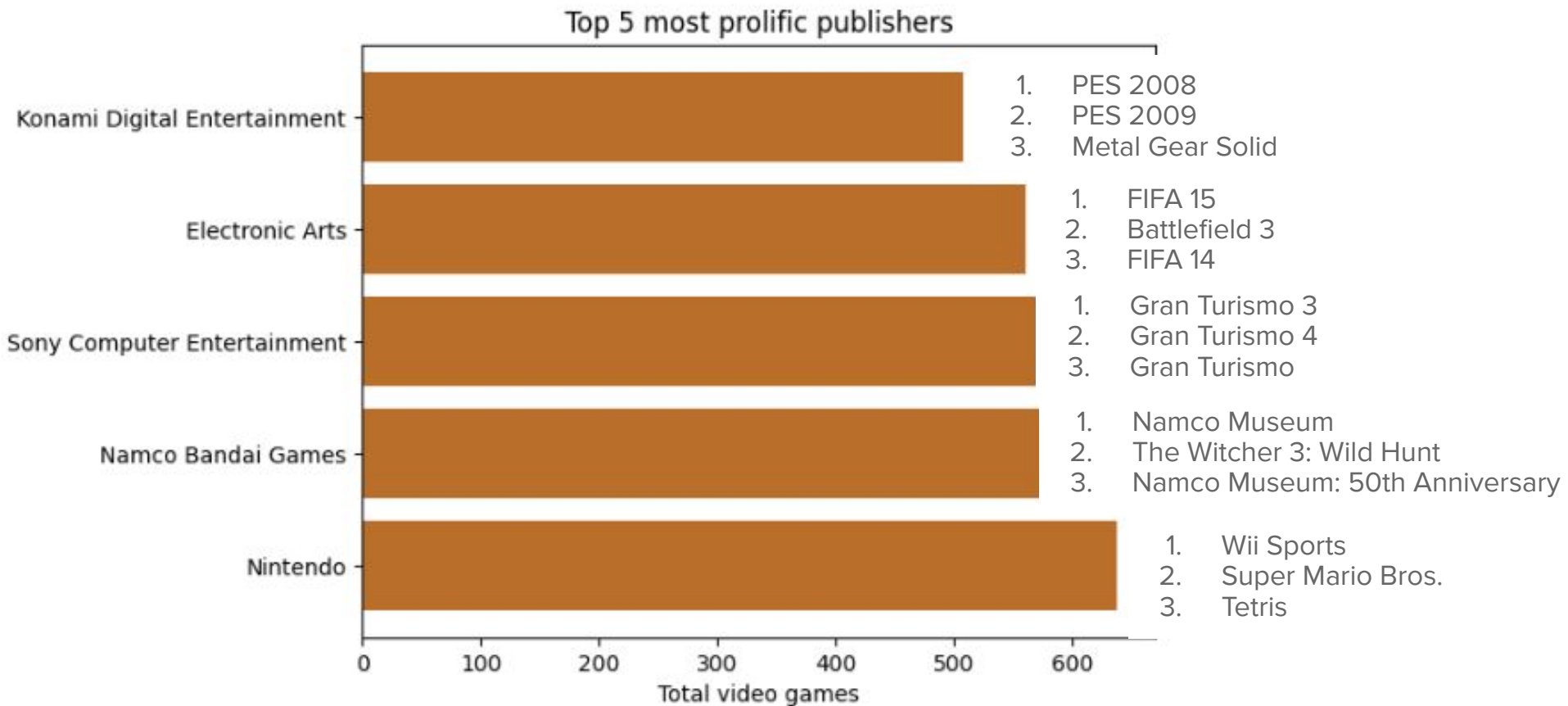
The number of yearly released video games was quite **stable until 1993** after then it gradually increased until reaching a **peak in 2008** with more than 1000 releases. A possible explanation for this is the release of the **same game on multiple platforms** at that time (*171 games were released on 2-7 platforms in 2008 compared to only 20 games released on 2-3 platforms in 2000*).

# Most prolific publishers



**Nintendo** was the most prolific publisher during that time with more than 600 video games released, followed by Namco Bandai, Sony, Electronic Arts & Konami.

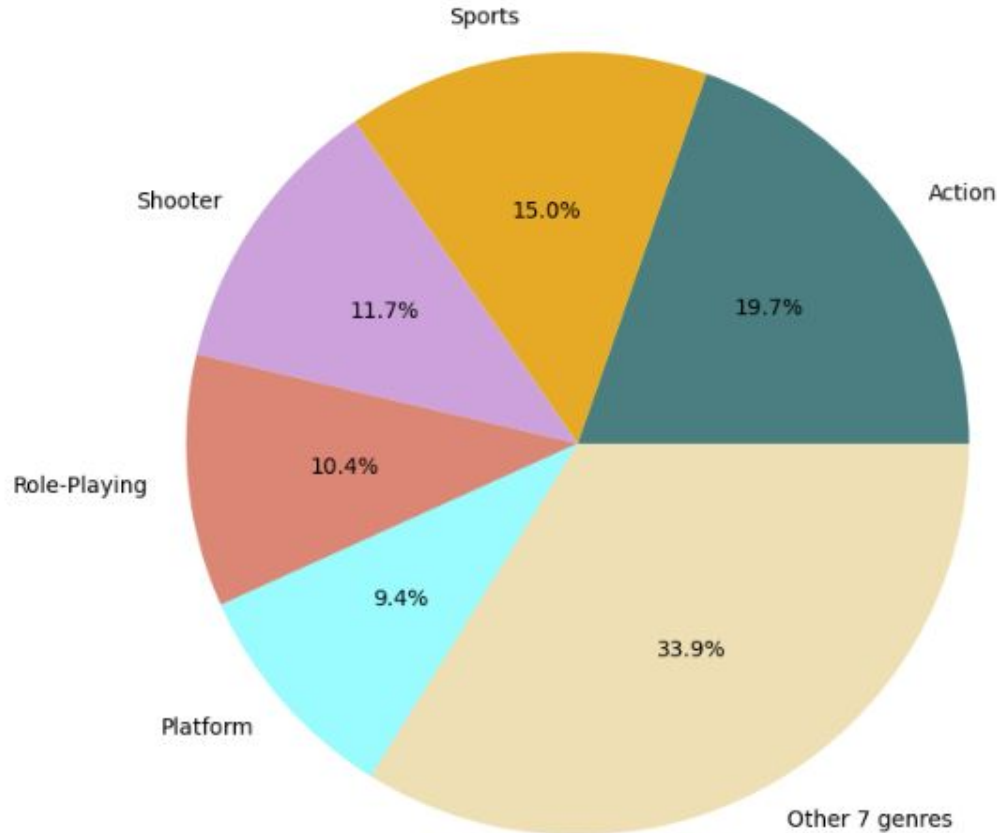
# Publishers' top 3 best-sellers



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# Best-selling genres globally



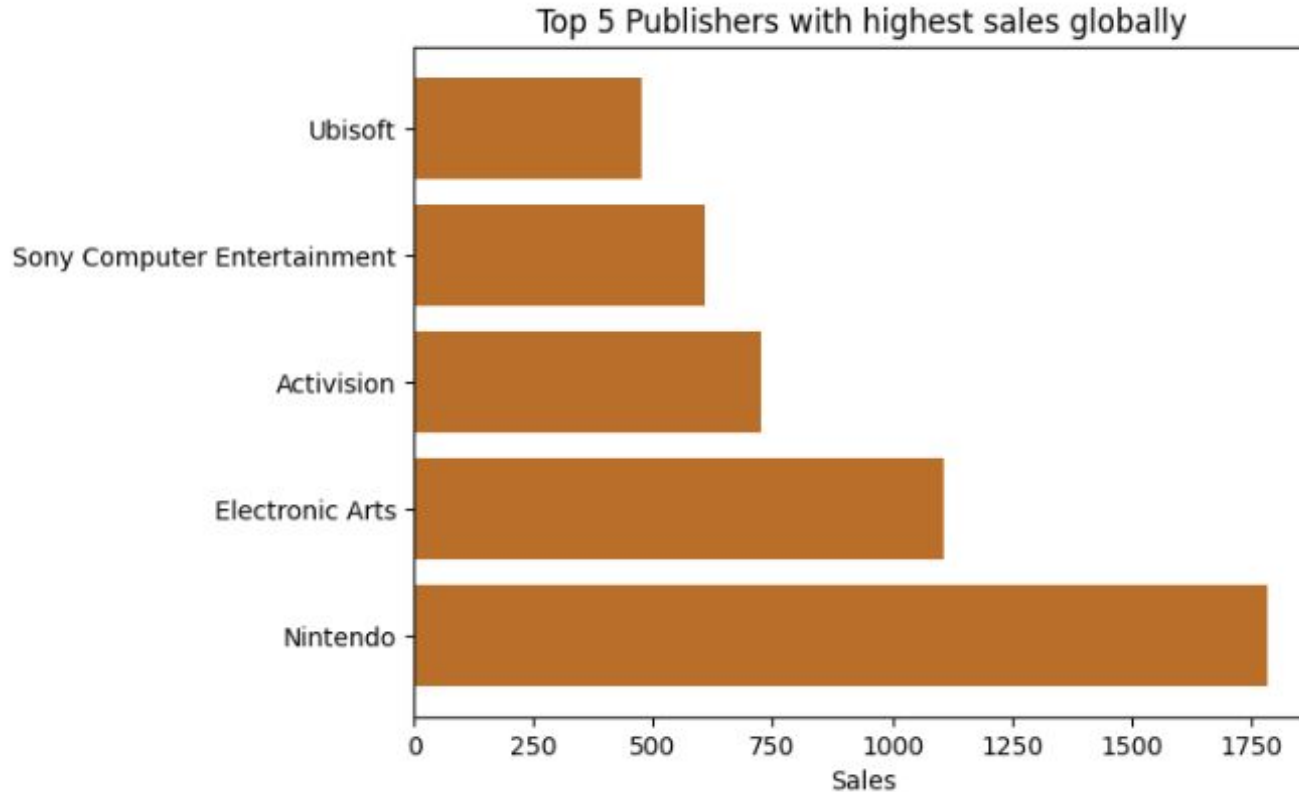
> Action, Sports & Shooter represented more than **46% of the global sales.**

# Best-selling game for each genre

| Genre        | Name                     | Global_Sales | Platform_count |
|--------------|--------------------------|--------------|----------------|
| Sports       | Wii Sports               | 82.74        | 1              |
| Action       | Grand Theft Auto V       | 55.92        | 5              |
| Platform     | Super Mario Bros.        | 45.31        | 2              |
| Puzzle       | Tetris                   | 35.84        | 2              |
| Racing       | Mario Kart Wii           | 35.82        | 1              |
| Role-Playing | Pokemon Red/Pokemon Blue | 31.37        | 1              |
| Shooter      | Call of Duty: Black Ops  | 31.03        | 5              |
| Misc         | Wii Play                 | 29.02        | 1              |
| Simulation   | Nintendogs               | 24.76        | 1              |
| Fighting     | Super Smash Bros. Brawl  | 13.04        | 1              |
| Adventure    | Assassin's Creed         | 11.30        | 3              |
| Strategy     | Pokemon Stadium          | 5.45         | 1              |

> The overall best-seller globally was **Wii Sports**, a sports video game released in 2006 on only **one platform**; whereas the second best-seller was **GTA V**, an action video game released for the first time in 2013 and on **five platforms** total.

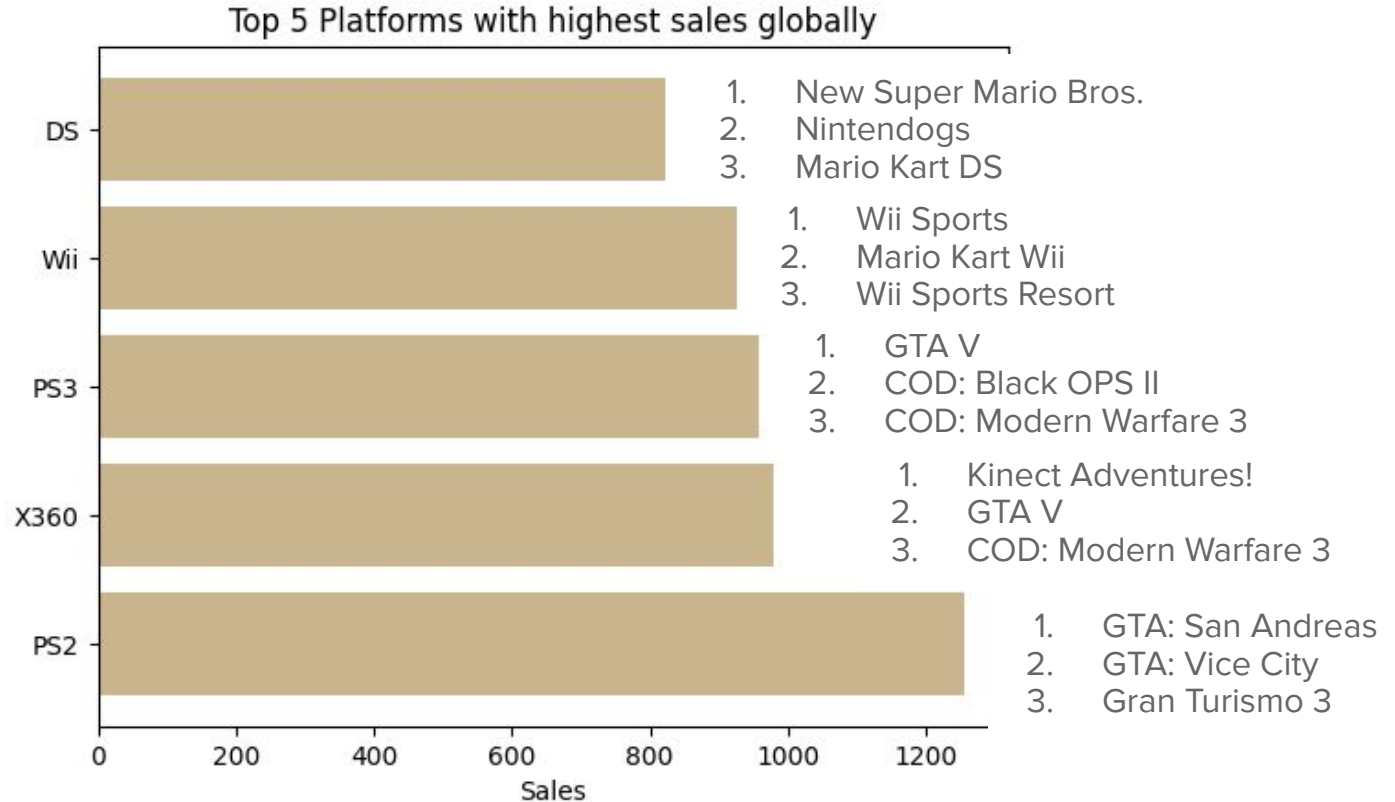
# Best-selling publishers



> **Nintendo** was by far the best-selling publisher with more than 1750 worth of sales globally.



# Best-selling platforms



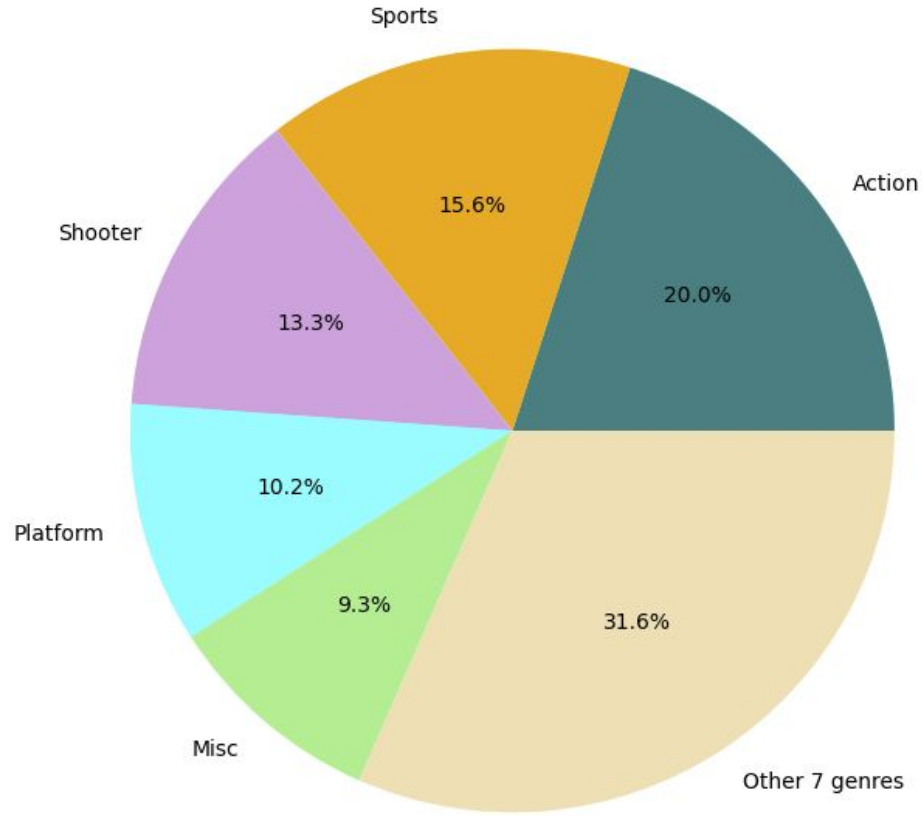
> **Playstation 2** was the most performant platform with more than 1200 worth of sales globally, followed by the XBOX360, Playstation 3 & Wii which all achieved similar sales figures.

> The **GTA & COD** franchises being pretty successful across multiple platforms.

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# Best-selling genres in North America



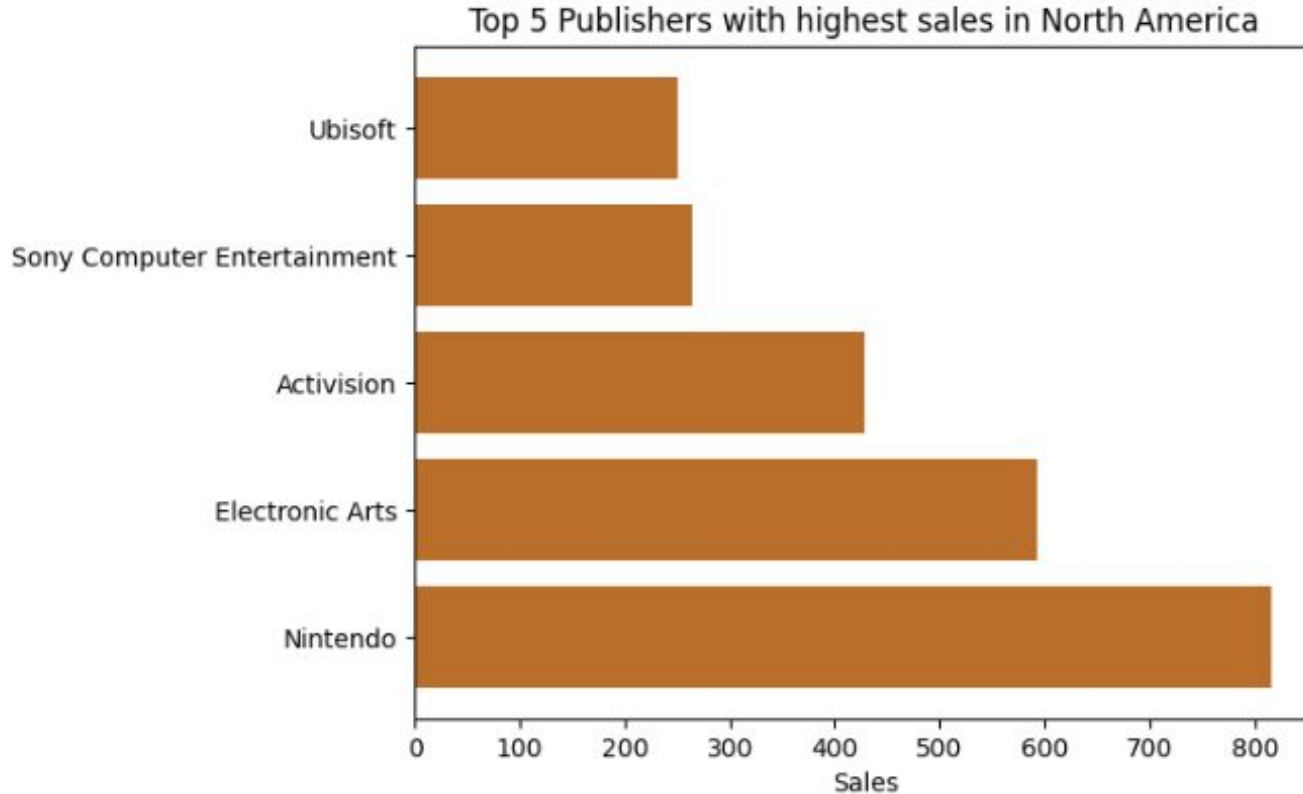
> Action, Sports & Shooter represented more than **48% of the sales in North America.**

# Best-selling game for each genre

| Genre        | Name                               | North_America_Sales | Platform_count |
|--------------|------------------------------------|---------------------|----------------|
| Sports       | Wii Sports                         | 41.49               | 1              |
| Platform     | Super Mario Bros.                  | 32.48               | 2              |
| Shooter      | Duck Hunt                          | 26.93               | 1              |
| Puzzle       | Tetris                             | 26.17               | 2              |
| Action       | Grand Theft Auto V                 | 23.46               | 5              |
| Racing       | Mario Kart Wii                     | 15.85               | 1              |
| Misc         | Kinect Adventures!                 | 14.97               | 1              |
| Role-Playing | Pokemon Red/Pokemon Blue           | 11.27               | 1              |
| Simulation   | Nintendogs                         | 9.07                | 1              |
| Fighting     | Super Smash Bros. Brawl            | 6.75                | 1              |
| Adventure    | Super Mario Land 2: 6 Golden Coins | 6.16                | 1              |
| Strategy     | Pokemon Stadium                    | 3.18                | 1              |

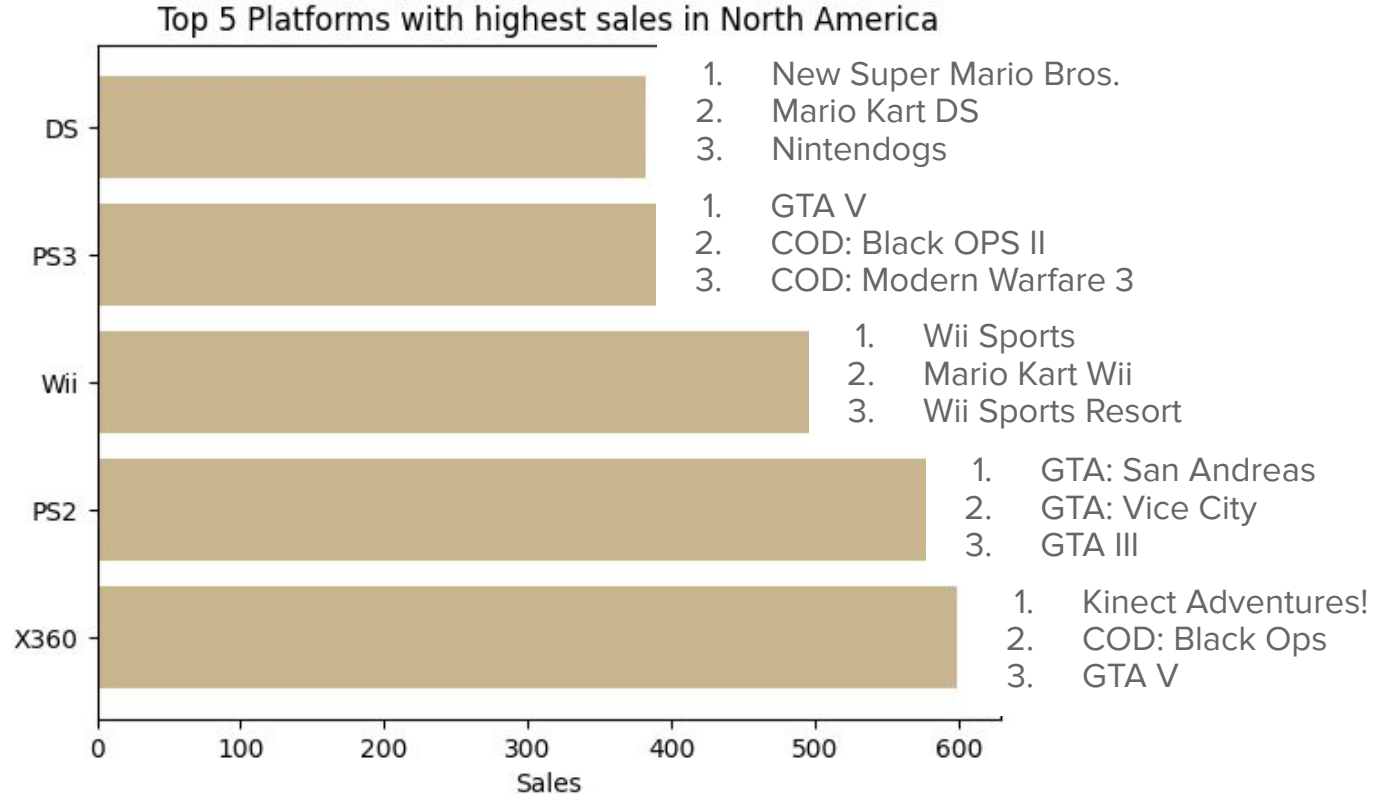
> The overall best-seller in North America was also **Wii Sports**, whereas the second best-seller was **Super Mario Bros.**, a platform video game first released in 1985 and on **two platforms** total.

# Best-selling publishers



> **Nintendo** was also the best-selling publisher in North America with more than 800 worth of sales. 21

# Best-selling platforms



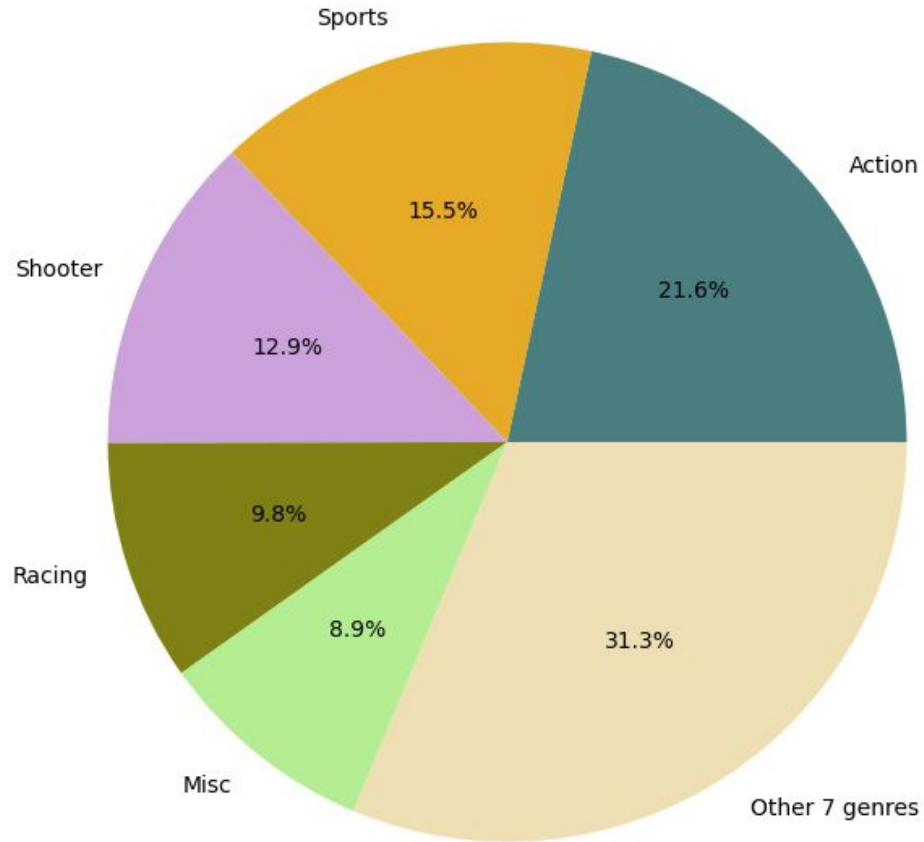
> In North America, the **XBOX360** was the platform with the highest sales, followed by the **Playstation 2** with similar performances.

> Again, the **GTA & COD** franchises being pretty successful across multiple platforms.

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# Best-selling genres in Europe



> Action, Sports & Shooter represented more than **50% of the sales in Europe.**

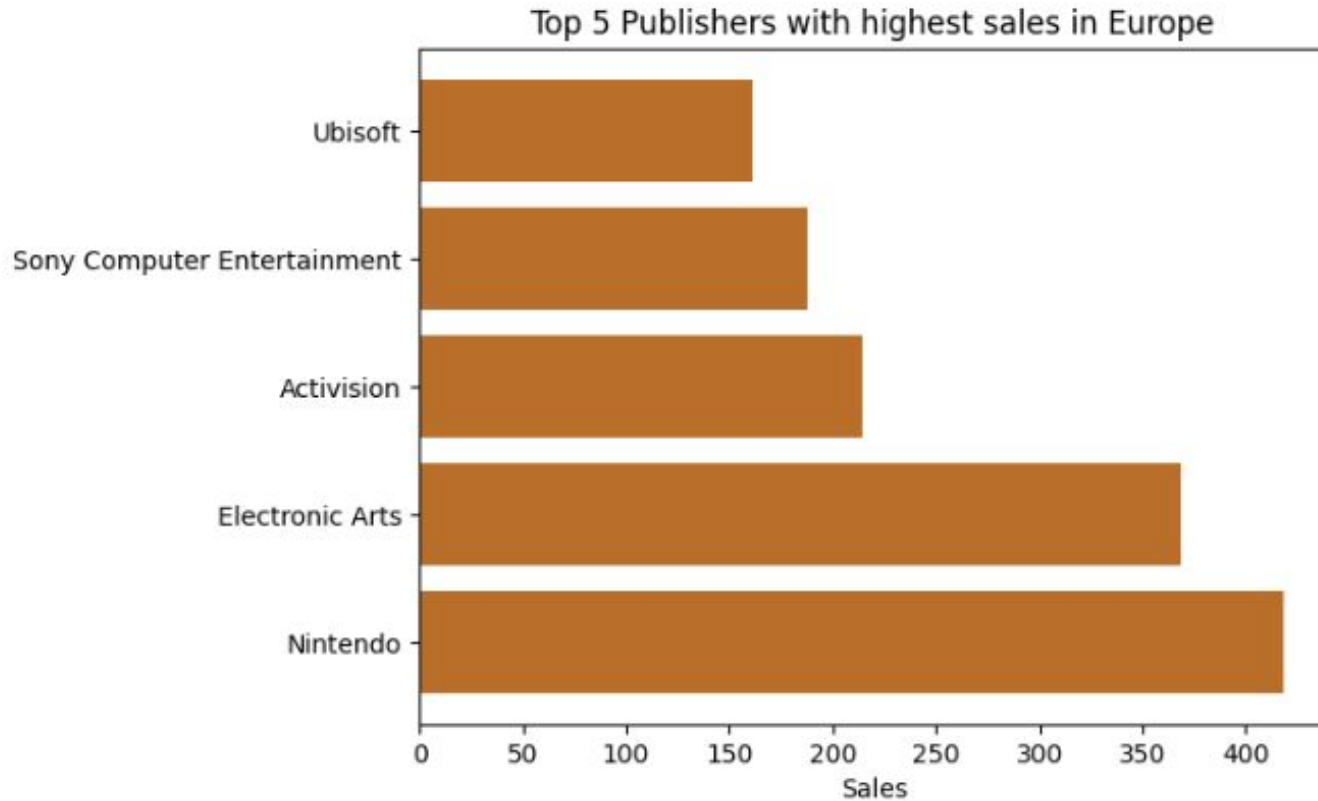


# Best-selling game for each genre

| Genre        | Name                                         | Europe_Sales | Platform_count |
|--------------|----------------------------------------------|--------------|----------------|
| Sports       | Wii Sports                                   | 29.02        | 1              |
| Action       | Grand Theft Auto V                           | 23.04        | 5              |
| Racing       | Mario Kart Wii                               | 12.88        | 1              |
| Shooter      | Call of Duty: Modern Warfare 3               | 11.29        | 4              |
| Simulation   | Nintendogs                                   | 11.00        | 1              |
| Misc         | Brain Age: Train Your Brain in Minutes a Day | 9.26         | 1              |
| Platform     | New Super Mario Bros.                        | 9.23         | 1              |
| Role-Playing | Pokemon Red/Pokemon Blue                     | 8.89         | 1              |
| Puzzle       | Brain Age 2: More Training in Minutes a Day  | 5.36         | 1              |
| Adventure    | Assassin's Creed                             | 4.48         | 3              |
| Fighting     | Super Smash Bros. Brawl                      | 2.61         | 1              |
| Strategy     | Warcraft II: Tides of Darkness               | 2.27         | 1              |

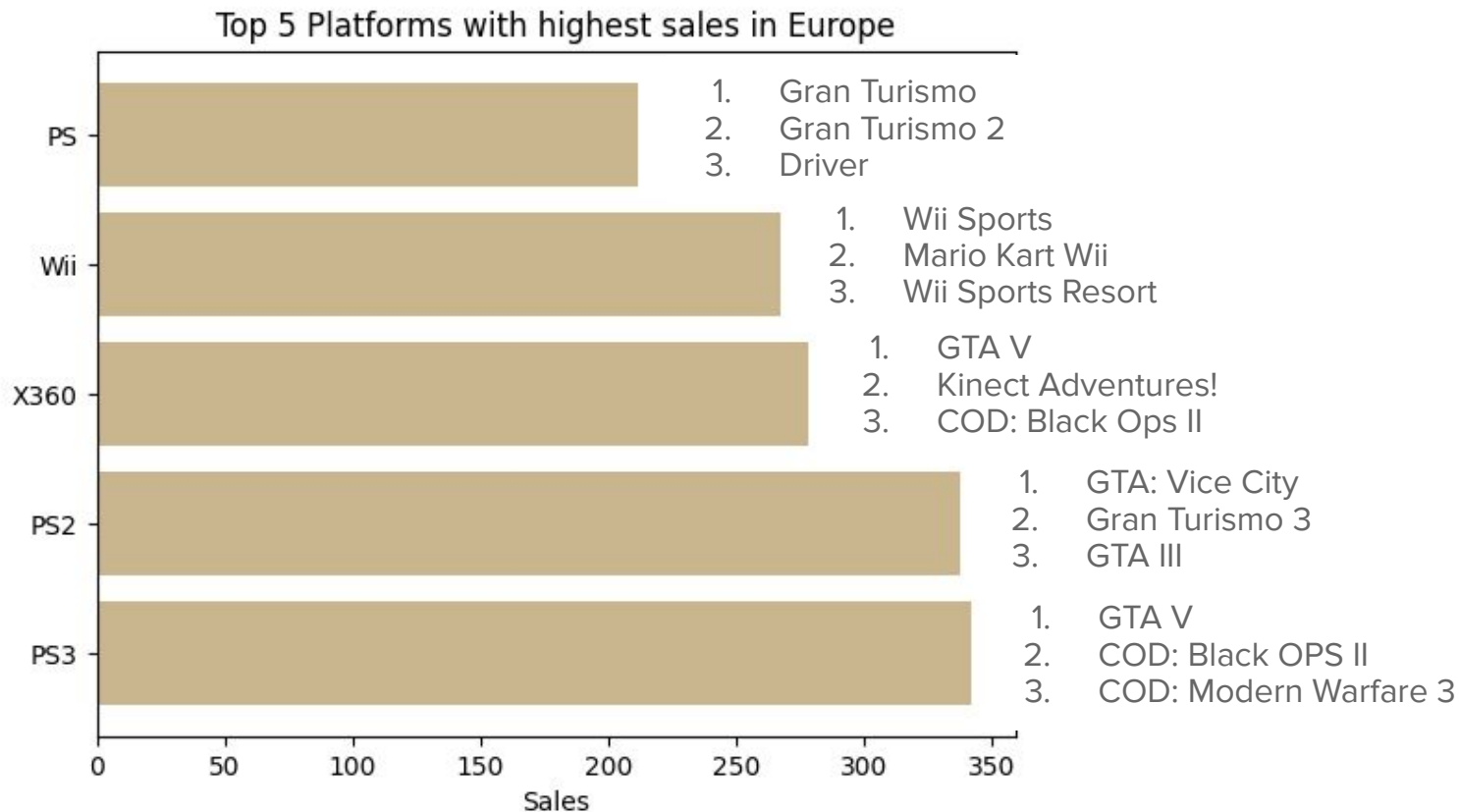
> The overall best-seller in Europe was also **Wii Sports**, followed by **GTA V**, as seen in global sales. 25

# Best-selling publishers



> **Nintendo** was, once again, the best-selling publisher with more than 400 worth of sales in Europe. 26

# Best-selling platforms



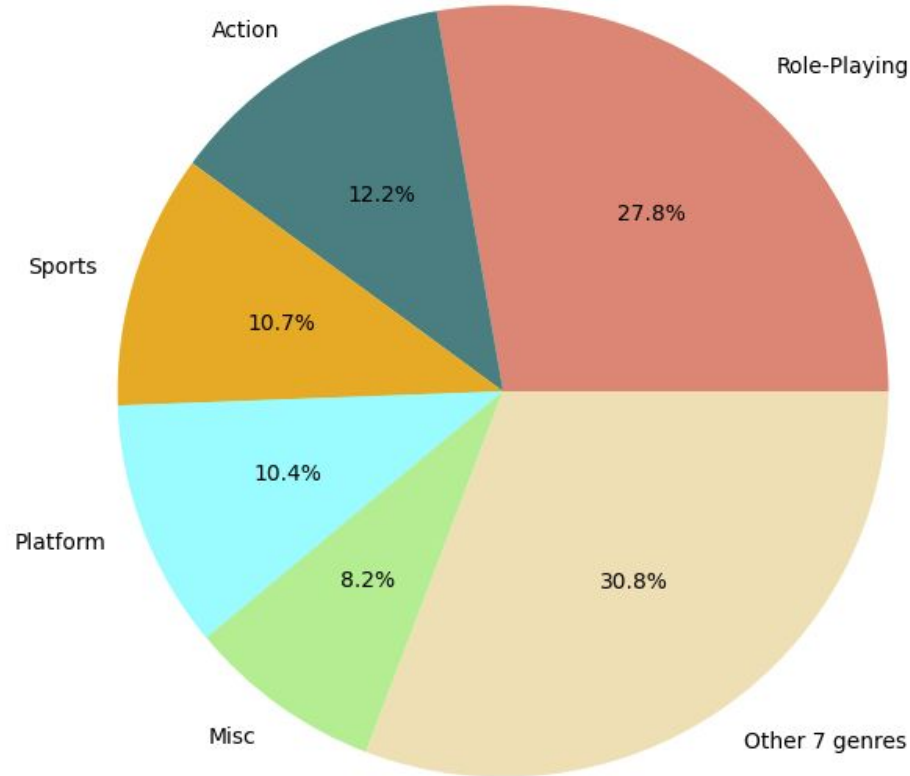
> In Europe, both **PlayStations 3 & 2** were the platforms with the highest sales, with similar performances.

> This time, on top of **GTA & COD** is **Gran Turismo**, another best-selling franchise across multiple platforms. <sup>27</sup>

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# Best-selling genres in Japan



> Role-Playing, Action & Sports represented more than **50% of the sales in Japan.**

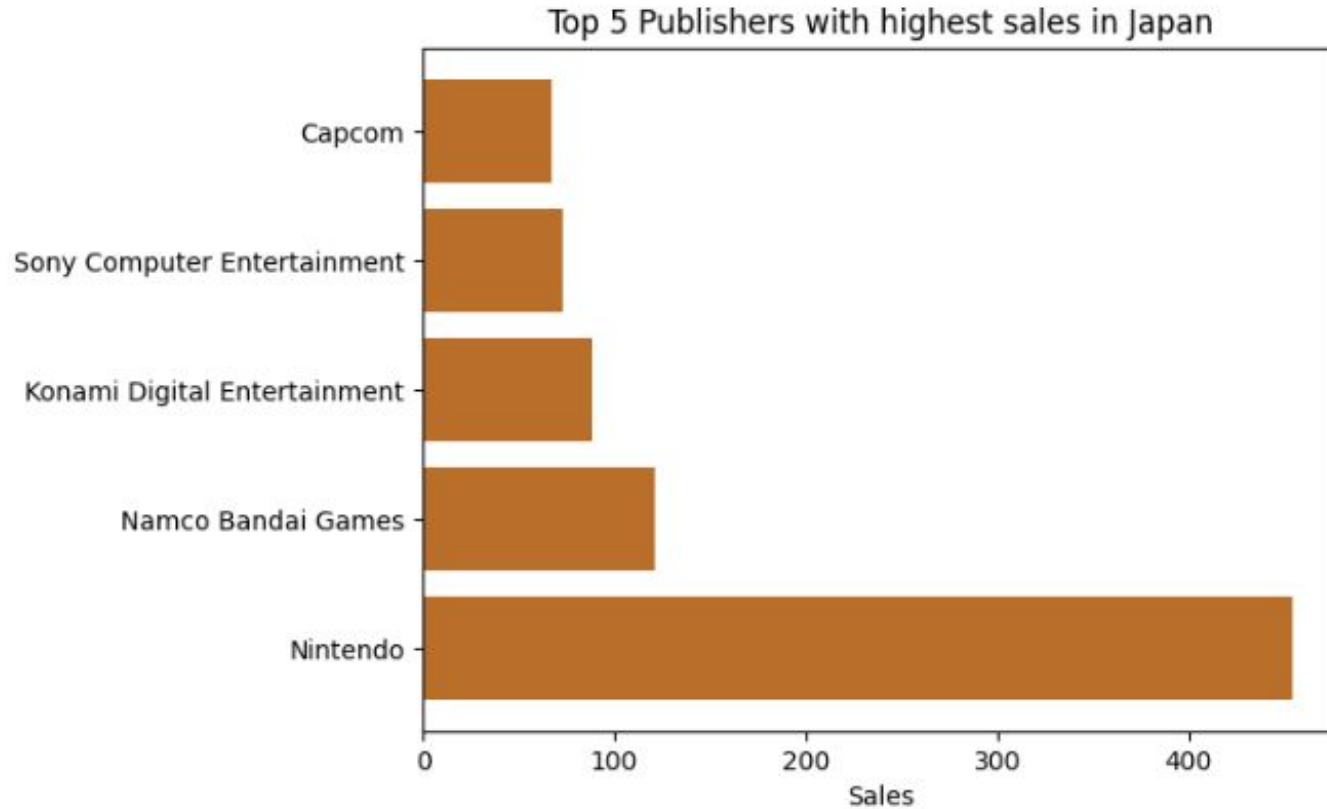
# Best-selling game for each genre

| Genre        | Name                                         | Japan_Sales | Platform_count |
|--------------|----------------------------------------------|-------------|----------------|
| Role-Playing | Pokemon Red/Pokemon Blue                     | 10.22       | 1              |
| Platform     | Super Mario Bros.                            | 6.96        | 2              |
| Puzzle       | Tetris                                       | 6.03        | 2              |
| Simulation   | Animal Crossing: Wild World                  | 5.33        | 1              |
| Misc         | Brain Age: Train Your Brain in Minutes a Day | 4.16        | 1              |
| Racing       | Mario Kart DS                                | 4.13        | 1              |
| Action       | Pokemon HeartGold/Pokemon SoulSilver         | 3.96        | 1              |
| Sports       | Wii Sports                                   | 3.77        | 1              |
| Fighting     | Super Smash Bros. for Wii U and 3DS          | 3.22        | 2              |
| Adventure    | Super Mario Land 2: 6 Golden Coins           | 2.69        | 1              |
| Strategy     | Yu-Gi-Oh! Duel Monsters                      | 1.61        | 1              |
| Shooter      | Splatoon                                     | 1.44        | 1              |

> In Japan, the top 3 overall best-selling video games were all released before the 2000s:

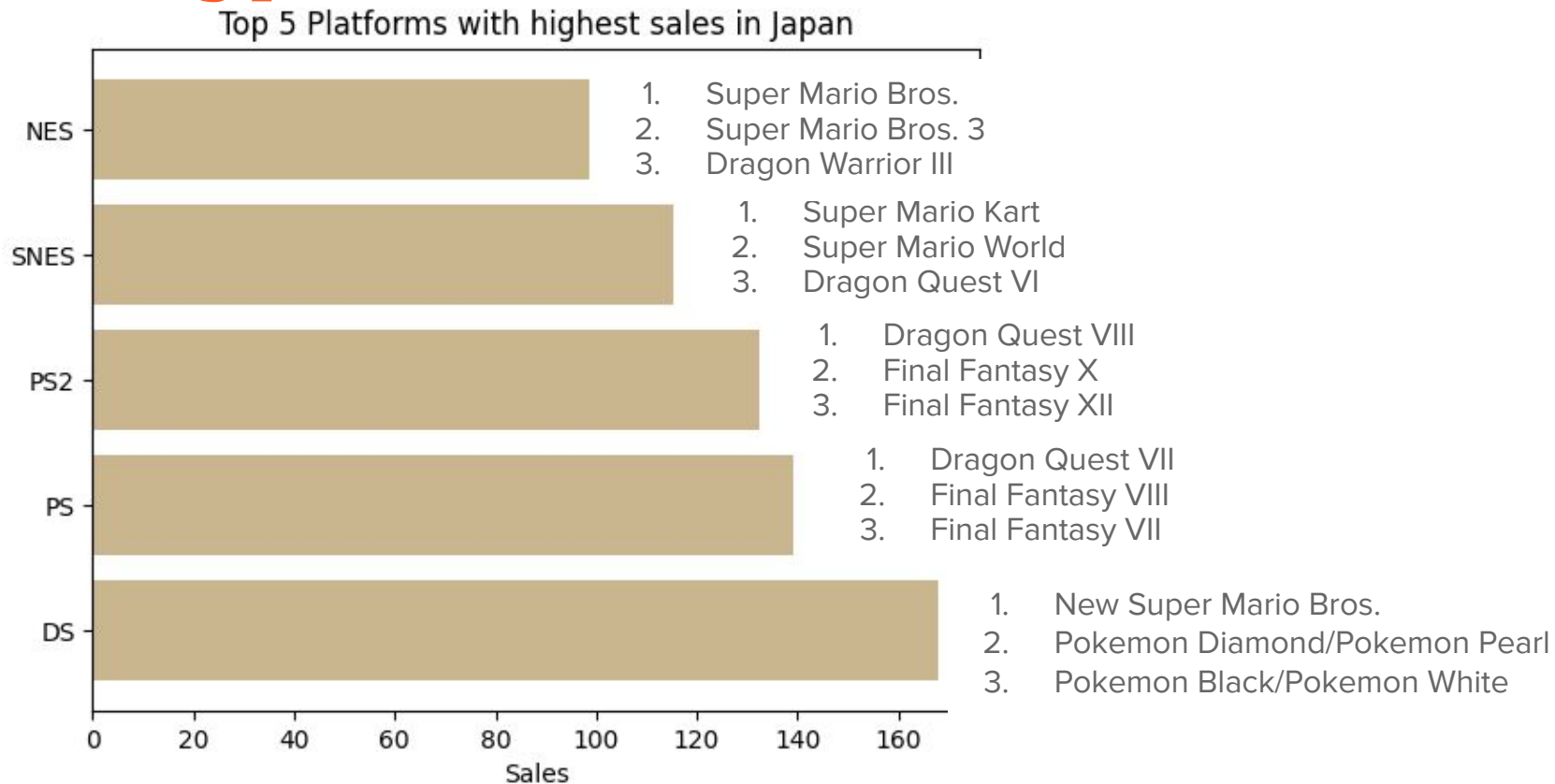
- **Pokemon Red/Pokemon Blue** released in 1996 on GameBoy
- **Super Mario Bros.** first released in 1985
- **Tetris** first released in 1988

# Best-selling publishers



> **Nintendo** was also the best-selling publisher with more than 400 worth of sales in Japan, way before the others with 4-time more sales overall.

# Best-selling platforms



> In Japan, the portable **Nintendo DS** was the platform with the highest sales, followed by the first two **PlayStations** and the classics NES & SNES.

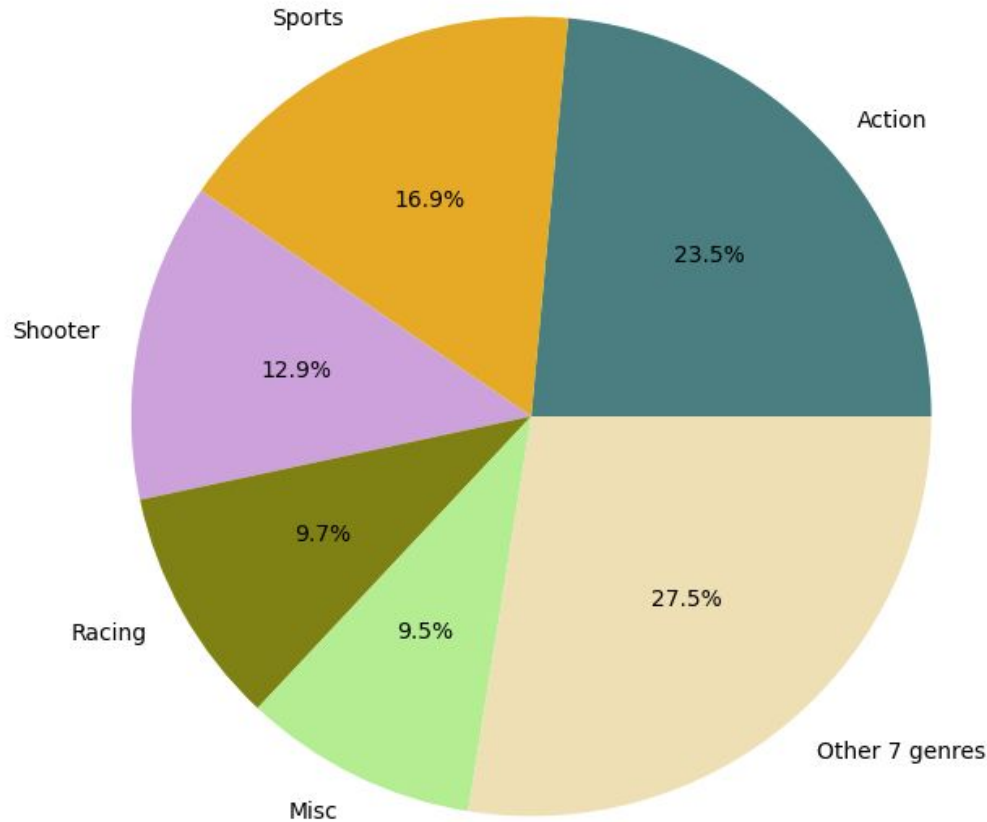
> Here, the **Final Fantasy**, **Dragon Quest** & **Mario** franchises have been pretty successful across platforms.



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# Best-selling genres in Other Regions



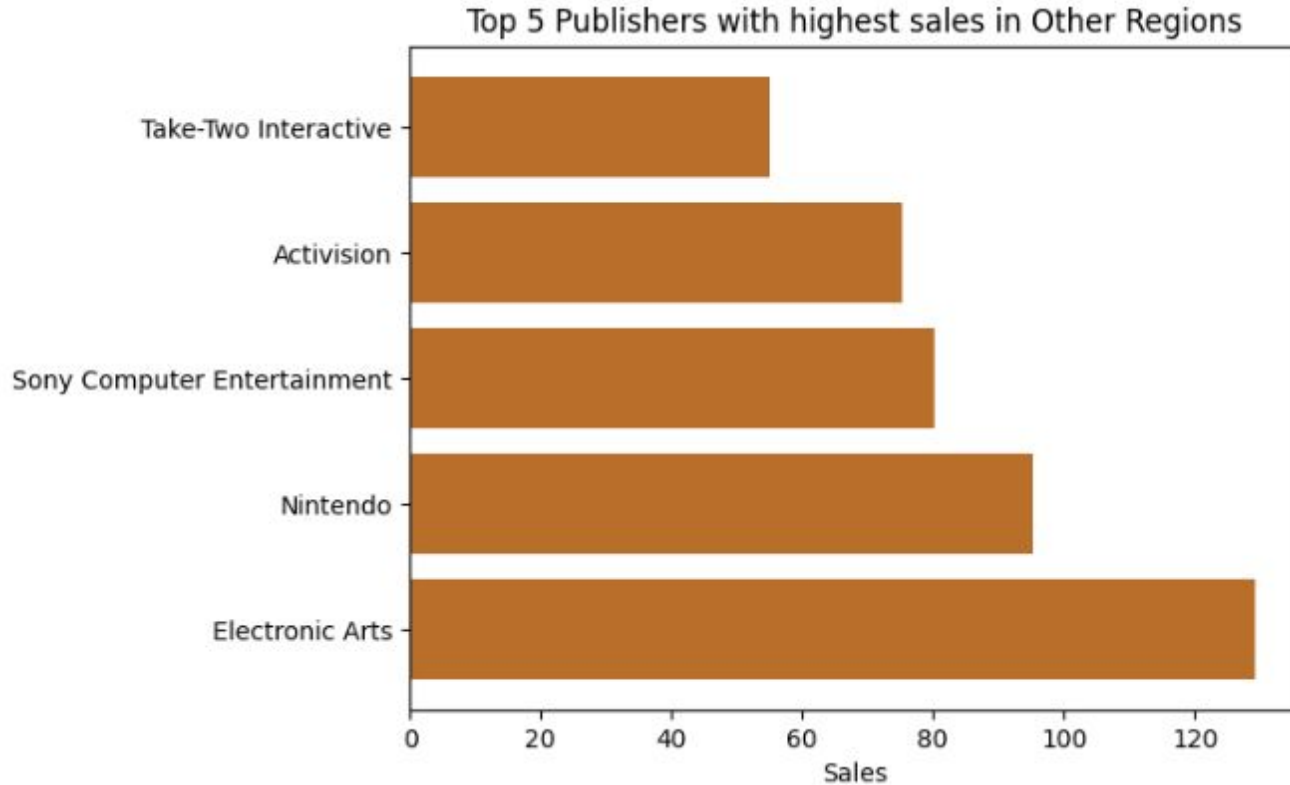
> Action, Sports & Shooter represented more than **53% of the sales in Other Regions.** 34

# Best-selling game for each genre

| Genre        | Name                                        | Other_regions_Sales | Platform_count |
|--------------|---------------------------------------------|---------------------|----------------|
| Action       | Grand Theft Auto: San Andreas               | 10.72               | 4              |
| Sports       | Wii Sports                                  | 8.46                | 1              |
| Racing       | Gran Turismo 4                              | 7.53                | 1              |
| Shooter      | Call of Duty: Black Ops II                  | 3.88                | 4              |
| Misc         | Minecraft                                   | 2.93                | 6              |
| Platform     | New Super Mario Bros.                       | 2.90                | 1              |
| Simulation   | Nintendogs                                  | 2.75                | 1              |
| Role-Playing | The Elder Scrolls V: Skyrim                 | 2.54                | 3              |
| Fighting     | WWE SmackDown vs Raw 2008                   | 2.15                | 6              |
| Adventure    | Assassin's Creed                            | 1.46                | 3              |
| Puzzle       | Brain Age 2: More Training in Minutes a Day | 1.18                | 1              |
| Strategy     | StarCraft II: Wings of Liberty              | 0.59                | 1              |

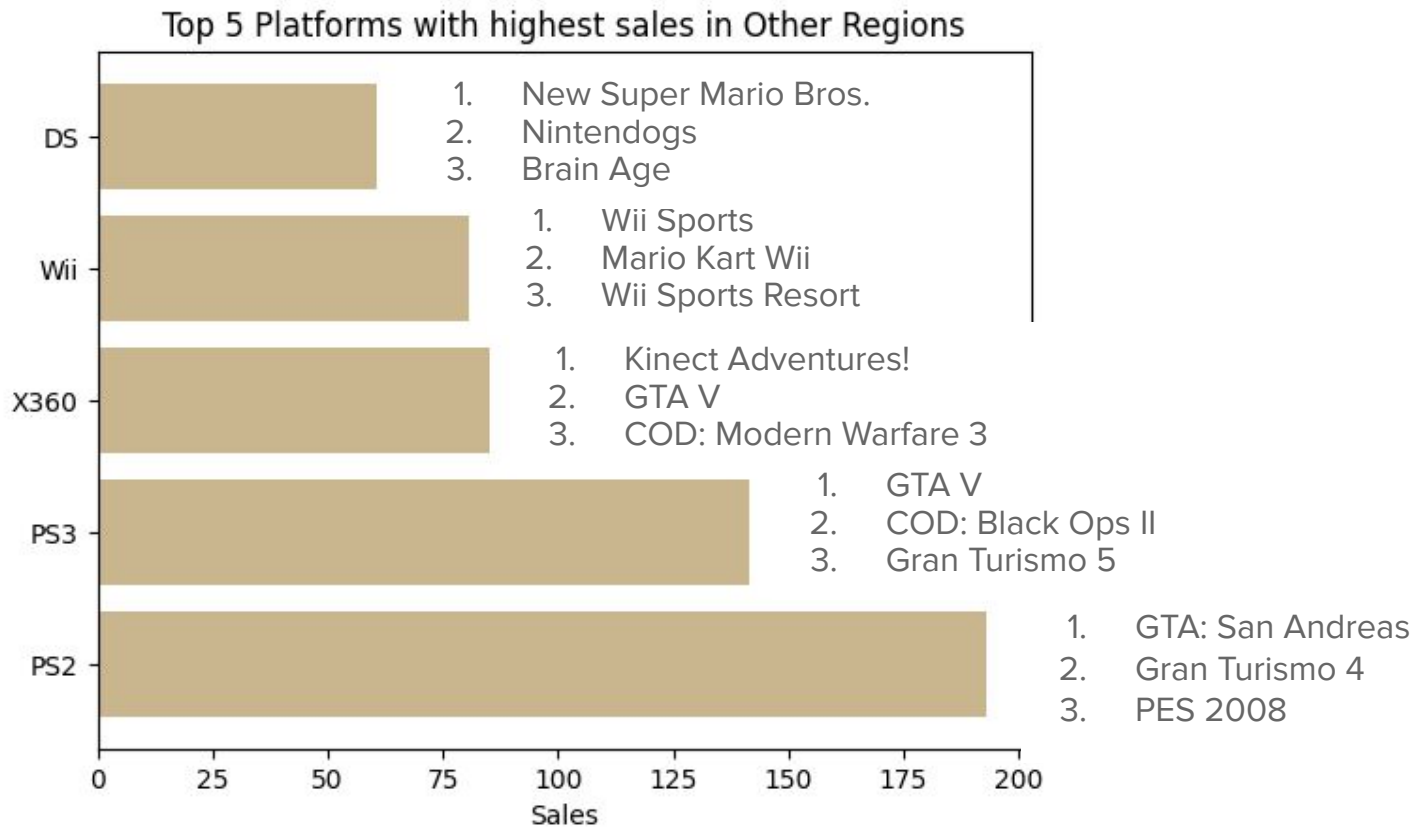
> The overall best-seller in Other Regions was **GTA San Andreas**, an action game first released in 2004, followed by **Wii Sports**, the best-seller video game in both North America & Europe.

# Best-selling publishers



> In Other Regions, **Electronic Arts** took the lead as the best-selling publisher with more than 120 worth of sales, way before Nintendo & others.

# Best-selling platforms



> **The PlayStation 2** was the platform with the highest sales in Other Regions, closely followed by the **third one**.

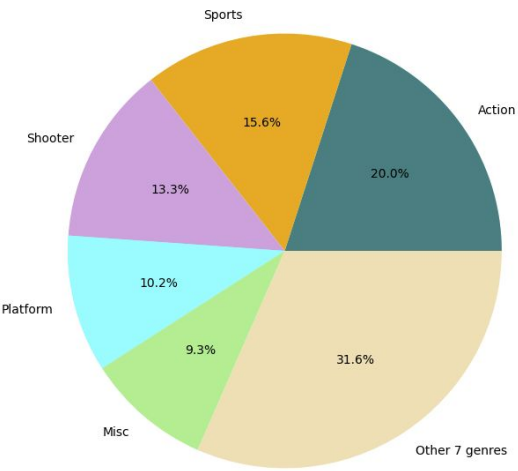
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# Comparison of the different regions

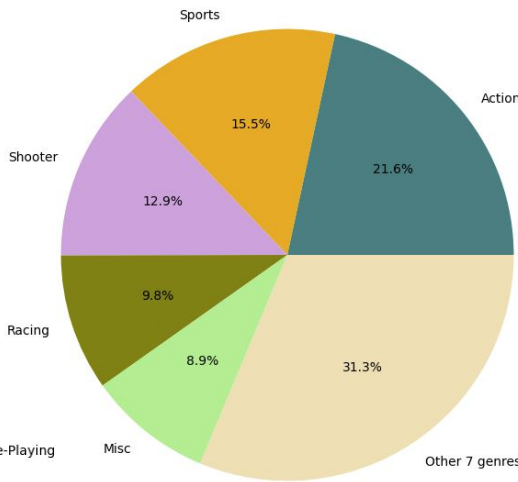
> **Europe & Other Regions** have the exact same taste in terms of genres with very similar sale performances

North America

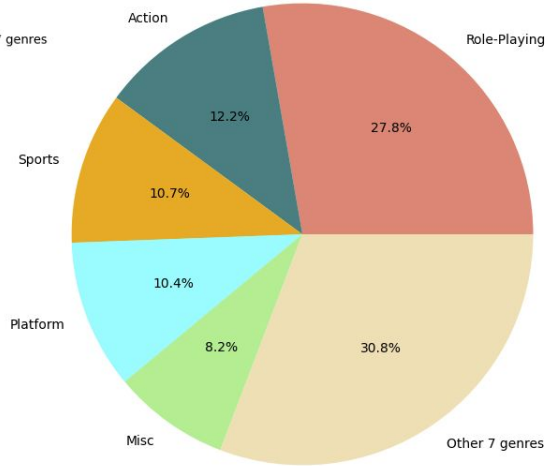


> **Action** & **Role-Playing** are the overall favorite genres

Europe

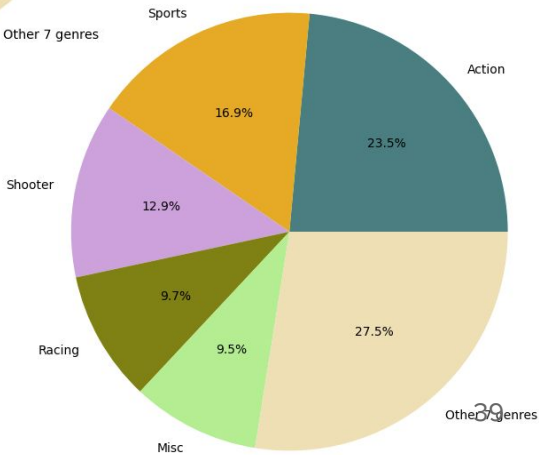


Japan



> **Japan** is the only region with a different Top 3 with **Role-Playing** being the best-selling genre

Other regions



# Comparison of the different regions

| Genre    | Name              | North_America_Sales | Platform_count |
|----------|-------------------|---------------------|----------------|
| Sports   | Wii Sports        | 41.49               | 1              |
| Platform | Super Mario Bros. | 32.48               | 2              |
| Shooter  | Duck Hunt         | 26.93               | 1              |

| Genre  | Name               | Europe_Sales | Platform_count |
|--------|--------------------|--------------|----------------|
| Sports | Wii Sports         | 29.02        | 1              |
| Action | Grand Theft Auto V | 23.04        | 5              |
| Racing | Mario Kart Wii     | 12.88        | 1              |

| Genre        | Name                     | Japan_Sales | Platform_count |
|--------------|--------------------------|-------------|----------------|
| Role-Playing | Pokemon Red/Pokemon Blue | 10.22       | 1              |
| Platform     | Super Mario Bros.        | 6.96        | 2              |
| Puzzle       | Tetris                   | 6.03        | 2              |

| Genre  | Name                          | Other_regions_Sales | Platform_count |
|--------|-------------------------------|---------------------|----------------|
| Action | Grand Theft Auto: San Andreas | 10.72               | 4              |
| Sports | Wii Sports                    | 8.46                | 1              |
| Racing | Gran Turismo 4                | 7.53                | 1              |

> All regions have different best-selling games but **Wii Sports** is first in both North America & Europe as well as second in Other Regions.

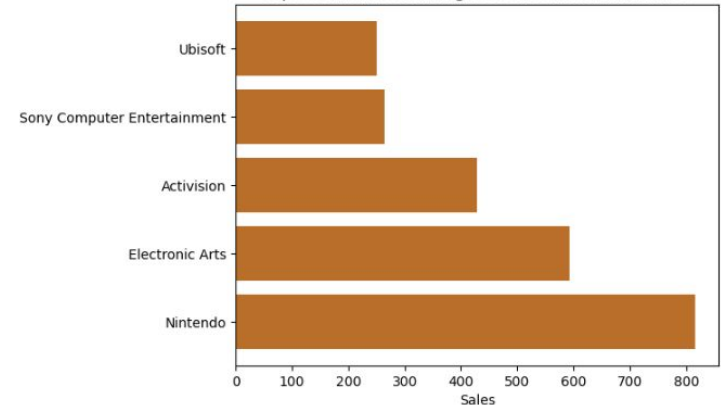
> The same franchise is present across different regions as well (GTA), & **Super Mario Bros.** is the second best-selling video games in both North America & Japan.

> It seems like the video games that have achieved the highest sales are not the ones that have been released on a multitude of platforms.

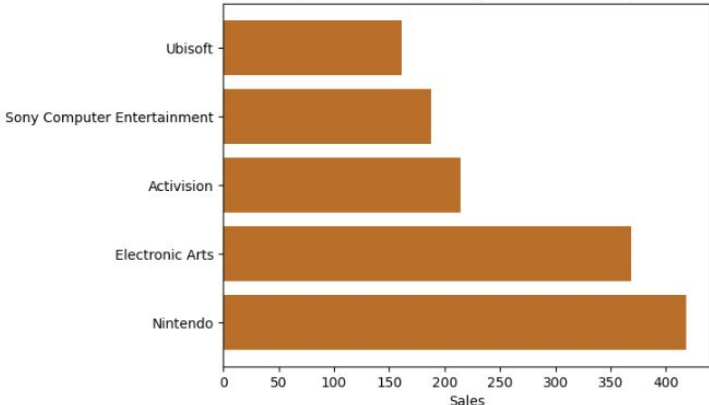


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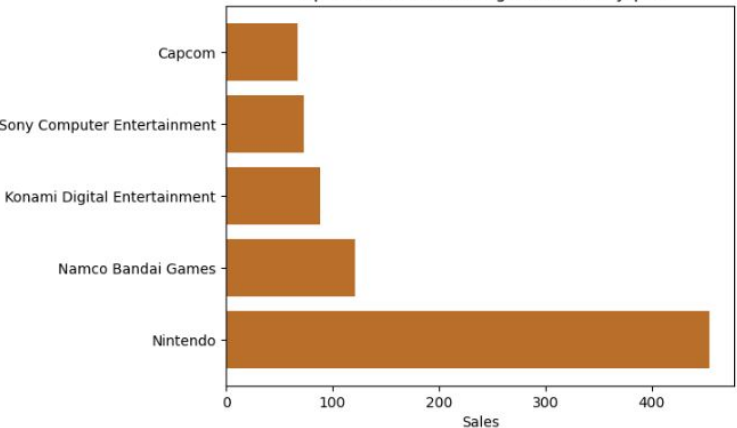
Top 5 Publishers with highest sales in North America



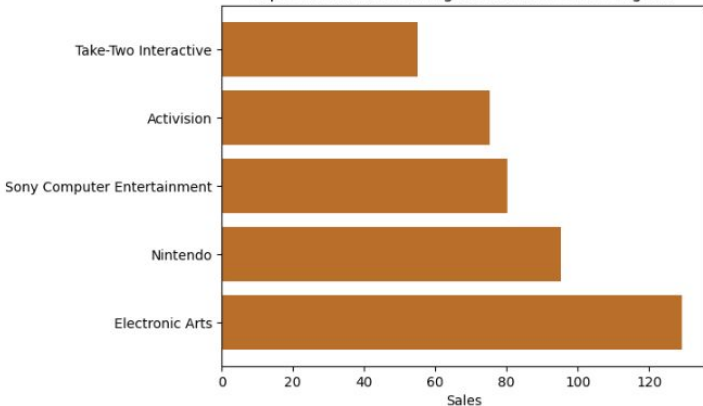
Top 5 Publishers with highest sales in Europe



Top 5 Publishers with highest sales in Japan

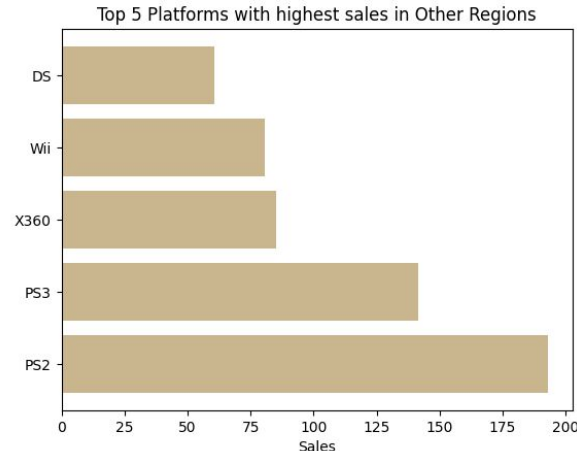
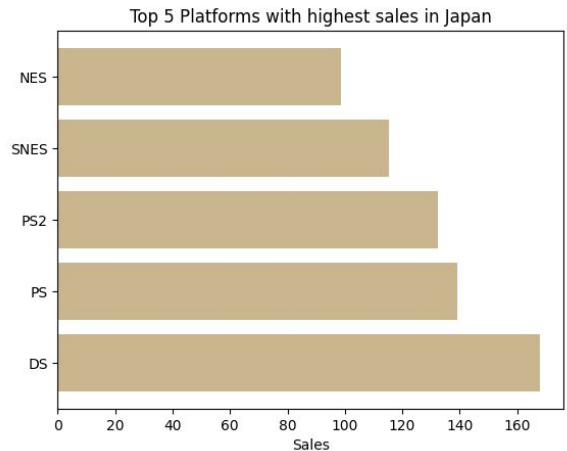
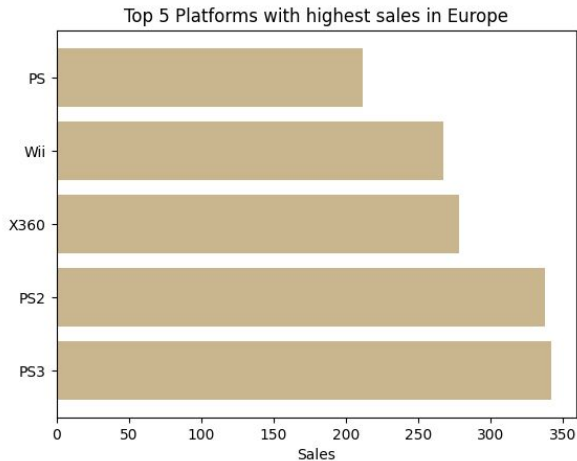
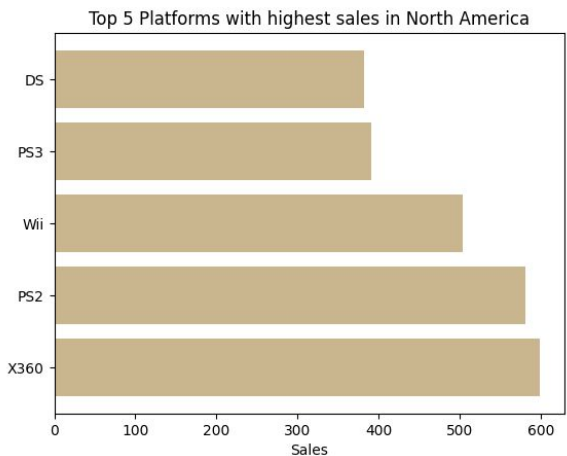


Top 5 Publishers with highest sales in Other Regions



> **Nintendo** is the most successful publisher everywhere except in Other Regions where **Electronic Arts** takes the lead

# Comparison of the different regions

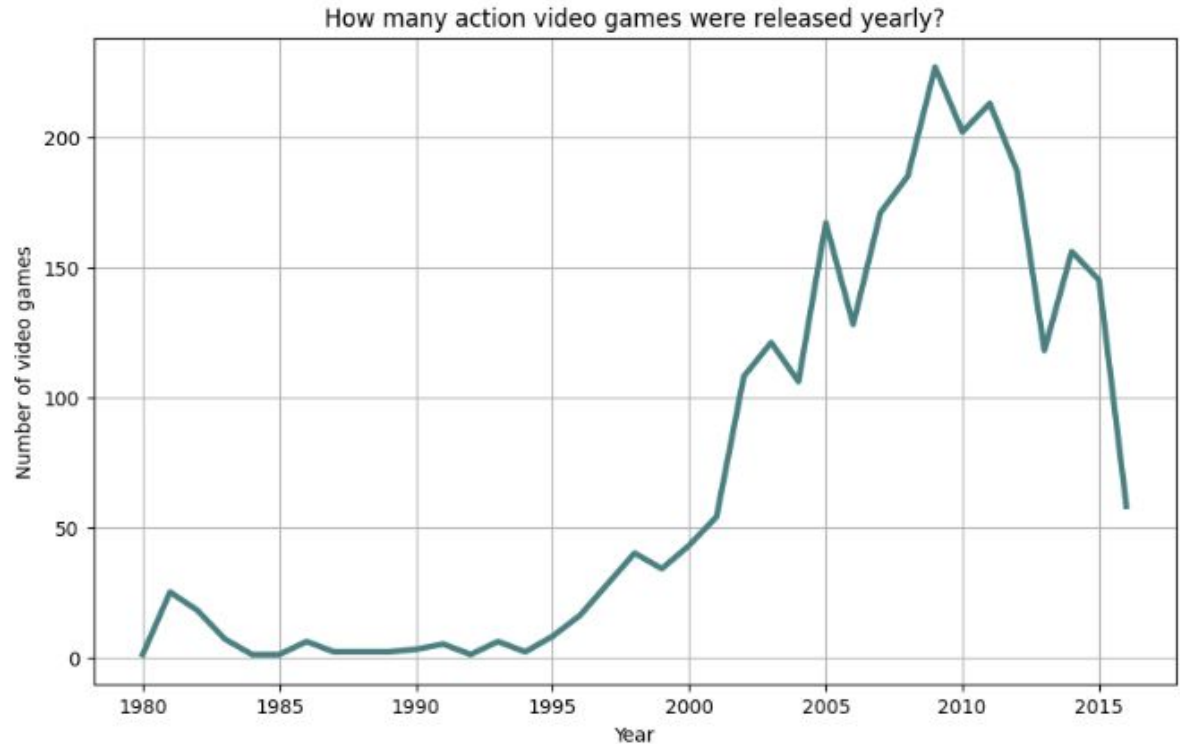
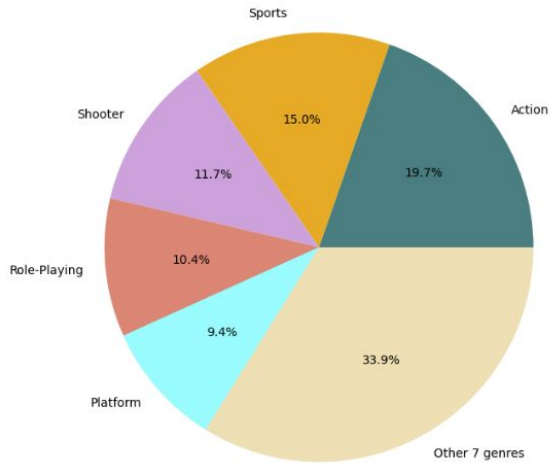


> **Japan** is the only region where a portable console is the best-selling platform  
> **Europe & Other Regions** both share a PlayStation as their best-seller while **North America** seems to prefer the XBOX360

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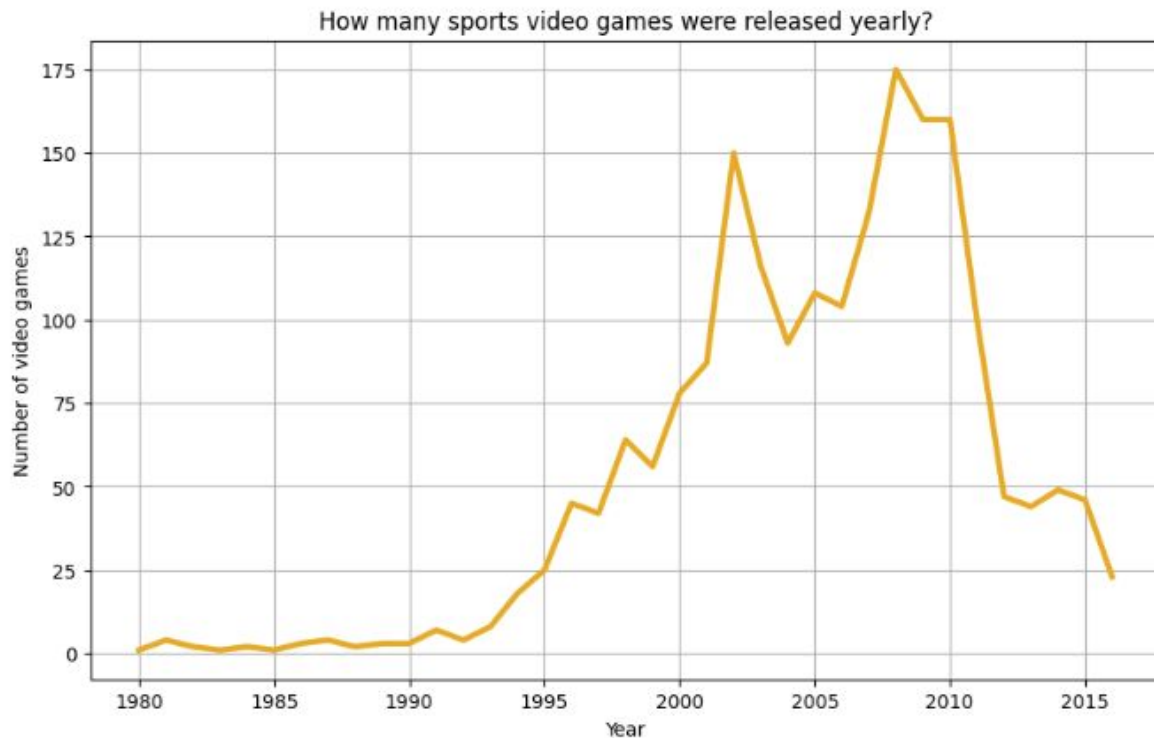
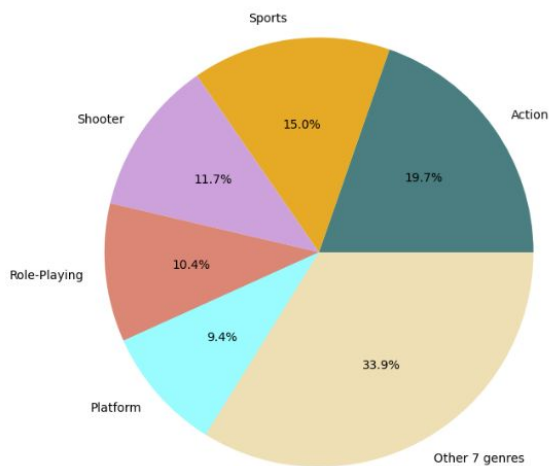
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# Action video games releases over time



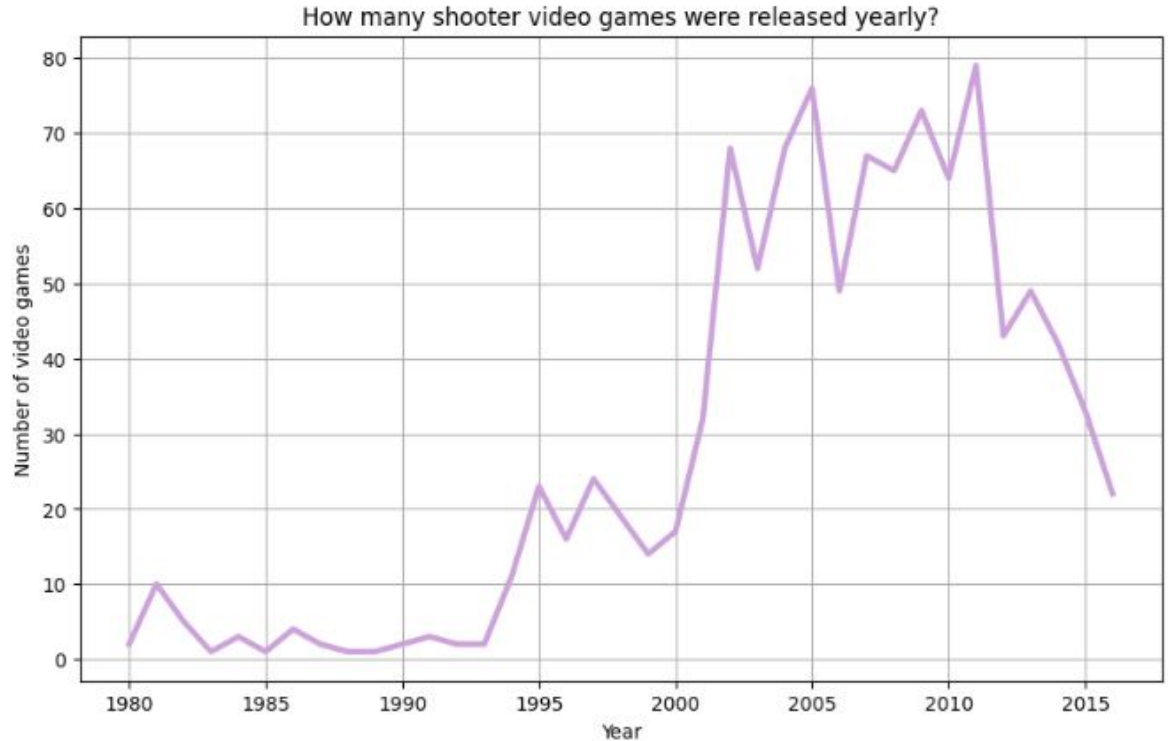
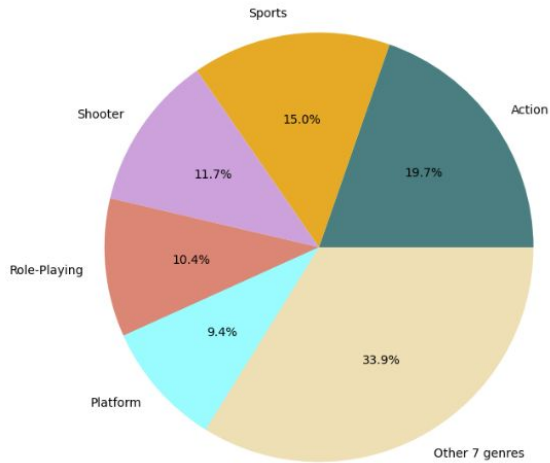
> The peak of releases for action video games was reached in **2009** with **Pokemon HeartGold/Pokemon SoulSilver** being the global best-seller that year.

# Sports video games releases over time



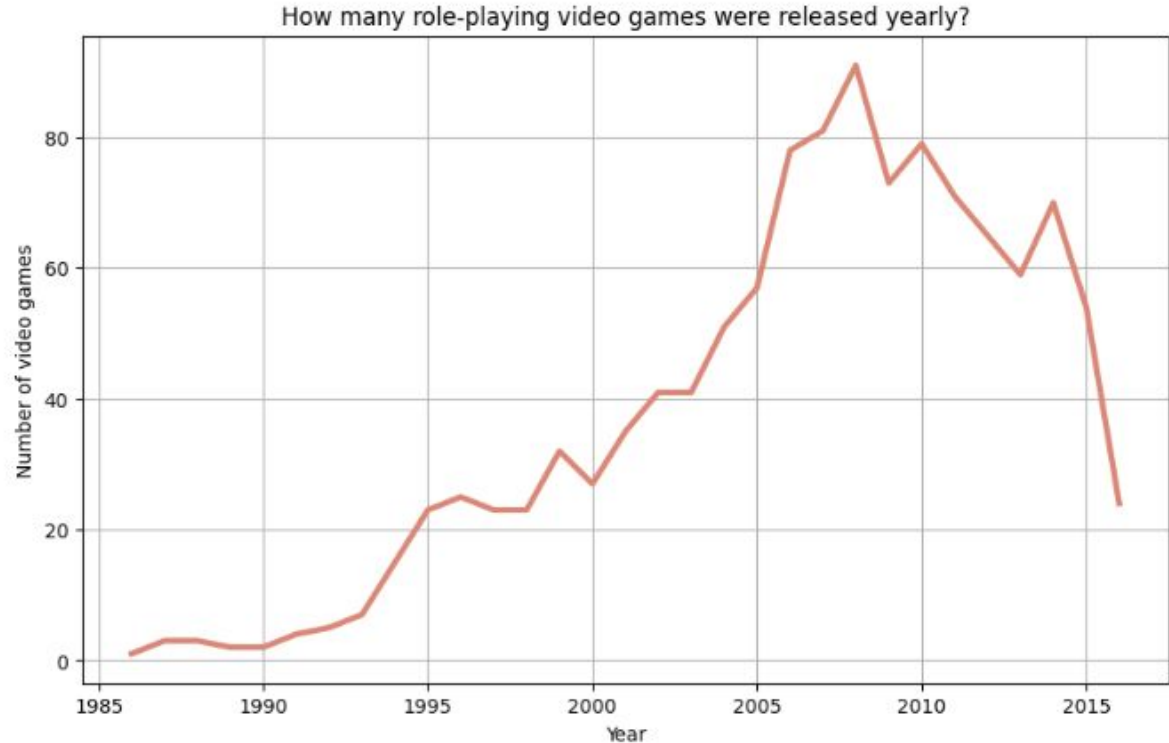
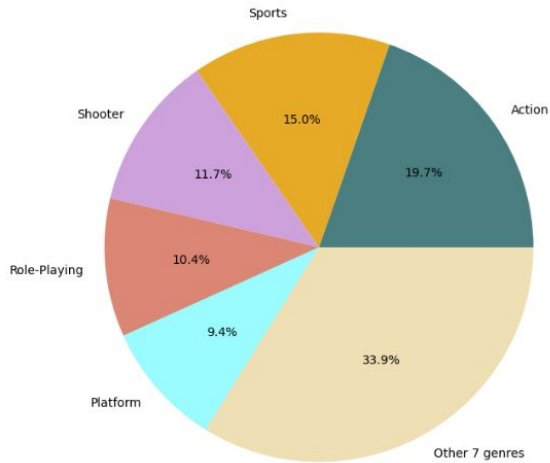
> The peak of releases for sports video games was reached in **2008** with **FIFA Soccer 09** being the global best-seller that year.

# Shooter video games releases over time



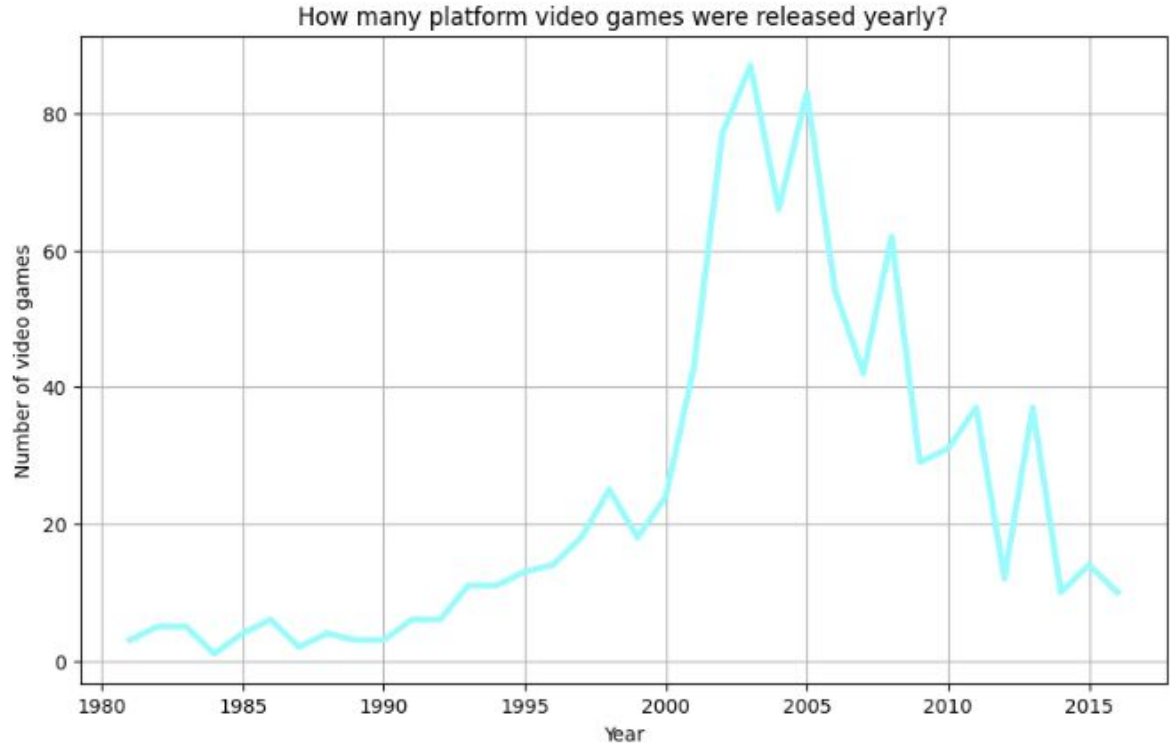
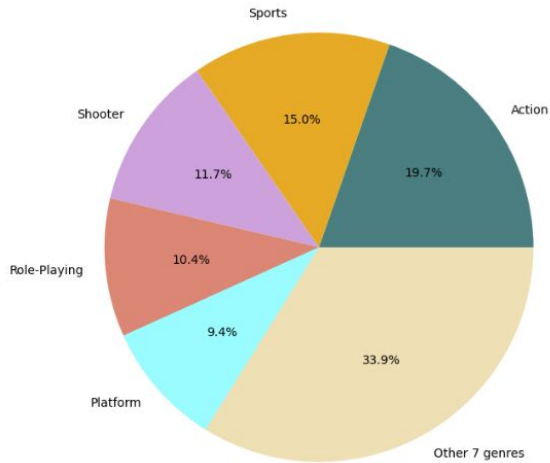
> The peak of releases for sports video games was reached in **2011** with **Call Of Duty: Modern Warfare 3** being the global best-seller that year.

# Role-playing video games releases over time



> The peak of releases for role-playing video games was reached in **2008** with **Fallout 3** being the global best-seller that year.

# Platform video games releases over time



> The peak of releases for platform video games was reached in **2003** with **Sonic Heroes** being the global best-seller that year.



# Content

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# Conclusion

The majority of video games released between 1980-2016 have been **action** (20.7%) followed by **sports** (15.5%) video games. This can explain why these genres have also been the highest selling ones globally.

Interestingly, the overall best-seller was a sports video game released in 2006 on only one platform: **Wii Sports on Wii**; whereas this was only the 4th most successful platform in terms of overall sales. Nonetheless, the Wii was developed by **Nintendo**, the most prolific & successful publisher globally.

If we take a closer look at the different regions, we can see that action was the best-selling genre everywhere except Japan, for which **role-playing** took the lead with almost 28% of the total sales. No wonder then that a role-playing game was Japan's best-seller as well: **Pokemon Red/Pokemon Blue**, whereas the other regions had either Wii Sports or the action game GTA: San Andreas. Regarding publishers, Nintendo was the favorite everywhere except in Other Regions where Electronic Arts made the highest sales. Finally, concerning platforms, only Japan had a portable console (Nintendo DS) as their most successful one.

If we take a closer look at the global best-selling genres, we can see a quite stable number of yearly releases until early to mid-1990s where there was a sudden increase until **a peak in mid to late-2000s** for all of them.